Predicting Powerlifting Total from Bodyweight, Age, Sex, and Equipment, with Secondary Analysis of DOTS

By Abdul Ahmad and Edwin David

1. Problem Formulation

Powerlifting is a strength sport in which athlete performance can be viewed in two related but different ways: absolute strength, such as total kilograms lifted in competition, and relative strength, such as DOTS, which adjusts performance across bodyweights. In this project, we focus primarily on predicting an athlete's competition TotalKg using a small set of athlete and competition characteristics, while using DOTS as a secondary analysis variable to interpret absolute versus relative performance. This framing is appropriate for a data science project because it involves a clear target variable, a small interpretable predictor set, and a real-world public dataset.

The main research question is: How accurately can an athlete's competition total be predicted using age, sex, equipment category and bodyweight? The secondary question is: How do these same variables relate to relative strength as measured by DOTS? These questions are specific, measurable, and aligned with the revised project plan documented in the repository notes.

This topic is also meaningful. Recent sport-science work has emphasized the growing research base around powerlifting and the need for evidence that can support athletes, coaches, and researchers. A recent scoping review noted an "exponential increase" in research on powerlifting and para powerlifting, while a recent prediction study in classic powerlifting modeled performance using variables such as "sex, age, weight" and competition history (Silverthorne et al., 2025). These findings support the relevance of modeling competition performance using interpretable athlete characteristics rather than purely black-box methods.

2. Problem Formulation

The dataset used in this project comes from the official OpenPowerlifting bulk CSV dataset, a public and widely used repository of competition results. OpenPowerlifting documents that its database is "distributed as a CSV file," and its bulk download page reports that the complete dataset contains roughly 3.89 million rows and is updated regularly (Open Powerlifting 2026). This makes it highly suitable for a public, reproducible project built around real-world data.

The raw file used in our project is openpowerlifting.csv, stored in data/raw/. The dataset summary generated from our code shows that the raw dataset had 3,891,860 rows and 42 columns at the time of analysis, and the required project columns were present which were Sex, Event, Equipment, Age, BodyweightKg, TotalKg, and DOTS and the raw dataset also included substantial amounts of missingness in some columns,

especially Age, which had 1,447,368 missing values before filtering meaning massive amounts of scrubbing had to be done in order to have only rows with all required data (Open Powerlifting 2026) (Ahmad & David 2026).

In each row in the dataset represents a competition result for a lifter and OpenPowerlifting documents several variables that are central to this project. Sex records the category in which the lifter competed (Men's category vs Womens category); Event records the competition type, where SBD indicates full powerlifting; Equipment records the equipment category such as Raw, Wraps, or Single-ply; Age records the age of the lifter if known; BodyweightKg records competition bodyweight; TotalKg is the sum of the three best successful lifts; and DOTS is a relative-strength score that can be calculated for valid performances. OpenPowerlifting also notes that TotalKg is empty if one of the lifts was failed or the lifter was disqualified, and that DOTS is intended to be compatible with older scoring systems while using updated coefficients.

Variables Chosen			
Variable	Type	What it represents	Why its Relevant
Sex	Categorical	The competition sex category of the lifter	Sex is a meaningful athlete characteristic and is associated with differences in absolute lifting performance, so it is appropriate as a predictor in the main regression model
Event	Categorical	The competition event type, such as full powerlifting (SBD)	This variable was mainly used for filtering the raw data to keep comparable full-powerlifting records rather than as a final predictor in the main model
Equipment	Categorical	The lifting equipment category used in competition, such as Raw, Wraps, Single-ply, Multi-ply, Unlimited, or Straps	Equipment category is directly related to competition conditions and can affect total lifted, so it is an appropriate predictor in the main model
Age	Numerical	The age of the lifter at the time of competition, when available	Age is conceptually relevant to athletic performance and was kept in the main model despite missingness because the remaining usable sample was still very large

BodyweightKG	Numerical	The lifter's bodyweight in kilograms at the competition	Bodyweight is directly related to strength performance and was one of the most important predictors in both eda and the final model
TotalKG	Numerical	The total weight lifted in competition, measured in kilograms	This is the main outcome variable in the project and serves as the dependent variable in the regression model
DOTS	Numerical	A relative-strength performance score used to compare lifters across bodyweights	DOTS was not used as the main target variable, but it was kept for secondary analysis to compare absolute strength and relative strength
Tested	Categorical/ Binary	Indicates whether the competition was drug tested	This variable was examined during data cleaning but not used in the final model because it had many missing values and was outside the simplified core predictor set
Division	Categorical	The competitive division of the lifter	This variable was available after cleaning, but it was not used in the final model because it added complexity and was not part of the core research question
WeightClassKG	Categorical / Numerical-ranges	The athlete's bodyweight class	This variable overlaps conceptually with bodyweight and was not used in the final model because BodyweightKg was kept as the more direct numerical predictor

Table 1. Key variables used or considered in the project

This table summarizes the most relevant variables from the OpenPowerlifting dataset and explains how each variable contributed to the filtering, analysis, or final model. It demonstrates that the dataset was well understood before modeling and supports the variable-selection decisions described later in the report.

The variables used in this project were chosen because they are directly relevant to the research questions as TotalKg serves as the dependent variable because it is the main direct measure of competition output and has no missing values after our cleaning step. BodyweightKg, Age, Sex, and Equipment were chosen as predictors because they are conceptually relevant to lifting performance and align with both the sport-science

literature and our project scope (Ferrari et al., 2025). DOTS was kept for secondary analysis only, because it is valuable for interpreting relative strength but is not the main regression target.

3. Data Cleaning and Preprocessing

Data cleaning was necessary before analysis because the raw dataset was much larger and more complex than what was needed for the final model. The CMPUT 195 project rubric explicitly expects duplicate handling, missing-value handling, dropping irrelevant variables, categorical encoding, numerical transformation, and a short justification of the preprocessing process. Our preprocessing pipeline was designed around those requirements and was implemented through separate scripts to make the workflow reproducible and easy to follow.

preprocessing pipeline/Steps taken from raw data to model-ready data					
Step	Dataset Stage	Rows	Columns	What was done	Why it matters
1	Raw dataset	3,891,860	42	Loaded the full OpenPowerlifting CSV and checked the required columns for the project	This confirmed that the dataset contained the variables needed for the research question and established the starting point for preprocessing
2	Initial project subset	3,891,860	10 (kept for project work)	Selected the columns Sex, Event, Equipment, Age, BodyweightKg, TotalKg, DOTS, Tested, Division, and WeightClassKg	This reduced the raw dataset to variables that were relevant for cleaning, EDA, variable selection, and modeling
3	Duplicate removal	3,278,070	10	Removed 613,790 duplicate rows	Duplicate observations can distort distributions, grouped summaries, and model fitting, so they needed to be removed

4	Validity and comparability filtering	2,147,935	10	Filtered out invalid or unusable rows and kept comparable competition records, including full-powerlifting records where appropriate	This improved data validity and made the final working dataset more suitable for meaningful comparison and modeling
5	Cleaned working dataset	2,147,935	10	Saved the cleaned dataset to data/processed/powerlifting_working.csv	This created a stable intermediate dataset for eda and later model preparation
6	Missing-value review for final model variables	2,147,935	5 main model columns considered	Reviewed missingness in TotalKg, BodyweightKg, Age, Sex, and Equipment	This showed that Age was the only main-model variable with substantial missingness and guided the final preprocessing decision
7	Age-based row removal for modeling	1,470,898	5 before encoding	Dropped rows with missing Age, removing 677,037 rows	Age was kept because it was conceptually relevant to the research question, and the remaining sample was still very large
8	Categorical encoding	1,470,898	expanded during encoding	One-hot encoded Sex and Equipment	Linear regression requires numerical inputs, so categorical predictors needed to be encoded properly
9	Numerical transformation	1,470,898	final 10 model-ready columns	Standardized BodyweightKg and Age using z-scores	Standardization placed numerical predictors on comparable scales and supported clearer coefficient interpretation
10	Model-ready dataset	1,470,898	10	Saved the final dataset to data/processed/powerlifting_model_ready.csv	This created the exact dataset used for train-test splitting and regression modeling

Table 2. This table summarizes the major cleaning and preprocessing decisions made in the project, including column selection, duplicate removal, validity filtering, handling of missing age values, categorical encoding, and

numerical standardization. It shows how the original public dataset was transformed into a reproducible modeling dataset

The first preprocessing step was to reduce the raw dataset to variables relevant to the project as we retained the columns Sex, Event, Equipment, Age, BodyweightKg, TotalKg, DOTS, Tested, Division, and WeightClassKg in the cleaned working dataset. We then removed duplicate rows and filtered out records with invalid or missing core lifting information. In particular, the cleaning step filtered to comparable competition records and kept only full powerlifting entries where appropriate. The resulting cleaned working dataset contained 2,147,935 rows and 10 columns, and 613,790 duplicate rows had been removed. After filtering, the remaining missingness included 677,037 missing values in Age, 432,042 in Tested, 2,780 in Division, and 17,954 in WeightClassKg, while Sex, Event, Equipment, BodyweightKg, TotalKg, and DOTS had no missing values.

At the variable-selection stage, we decided to use TotalKg as the main target and BodyweightKg, Age, Sex, and Equipment as the main predictors. DOTS was explicitly retained for EDA and interpretation only. Event was used mainly for filtering comparable rows rather than as a final predictor, and columns such as Tested, Division, and WeightClassKg were left out of the main model because they either introduced unnecessary complexity, had notable missingness, or overlapped conceptually with simpler predictors already in the model. This final structure was chosen to keep the model interpretable while still remaining faithful to the research question.

For final model preparation, the selected columns were reduced to TotalKg, BodyweightKg, Age, Sex, and Equipment. Because Age remained part of the intended research question and was conceptually relevant to performance, we chose to keep it in the model and drop rows where age was missing. This removed 677,037 rows, leaving a model-ready dataset of 1,470,898 rows. Although this is a substantial reduction, the remaining dataset is still extremely large, which makes the decision defensible. In the final report interpretation, this will be acknowledged as a limitation of the modeling procedure rather than hidden as if it had no cost.

The categorical variables Sex and Equipment were one-hot encoded for use in the regression model and the numerical predictors BodyweightKg and Age were standardized manually using z-scores. The means and standard deviations used for scaling were recorded by the model-preparation script as BodyweightKg mean=83.535, standard deviation=22.012; Age mean=28.541, standard deviation=12.042. Standardization was appropriate here because it places numerical predictors on a comparable scale and makes coefficient interpretation more stable in a multi-variable linear model.

Overall, the preprocessing procedure was appropriate for the project as it dealt with duplicates, handled missing values explicitly, removed unnecessary variables from the main model, encoded categorical predictors, and standardized numerical predictors. It also preserved a clear distinction between the main target (TotalKg) and the secondary analysis variable (DOTS), which helped keep the project conceptually coherent.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was used to understand the structure of the cleaned dataset, examine patterns in competition performance, and guide the modeling choices. The EDA script used the variables TotalKg, BodyweightKg, Age, Sex, Equipment, and DOTS, and for age-based summaries it dropped rows with missing age, leaving 1,470,898 rows in the EDA subset. EDA should be used to examine structure, distributions, relationships, outliers, and potential variables of interest before modeling, and this is exactly how it was used here.

4.1 Descriptive statistics

Descriptive statistics for the main numerical variables used in the eda								
Variable	Count	Mean	Std Dev	Minimum	25th Percentile	Median	75th Percentile	Maximum
TotalKg	1,470,898	493.993	172.113	7.000	352.500	495.000	615.000	1407.500
BodyweightKg	1,470,898	83.535	22.012	16.950	67.050	81.470	97.310	285.000
Age	1,470,898	28.541	12.042	0.000	19.500	25.000	34.500	95.500
DOTS	1,470,898	363.436	80.072	8.490	310.720	362.320	414.610	808.170

Table 3. This table summarizes the central tendency and spread of the key numerical variables after rows with missing age were removed for age-based analysis. It provides an overall view of the scale and variability of competition totals, bodyweight, age, and DOTS before modeling and interpretation

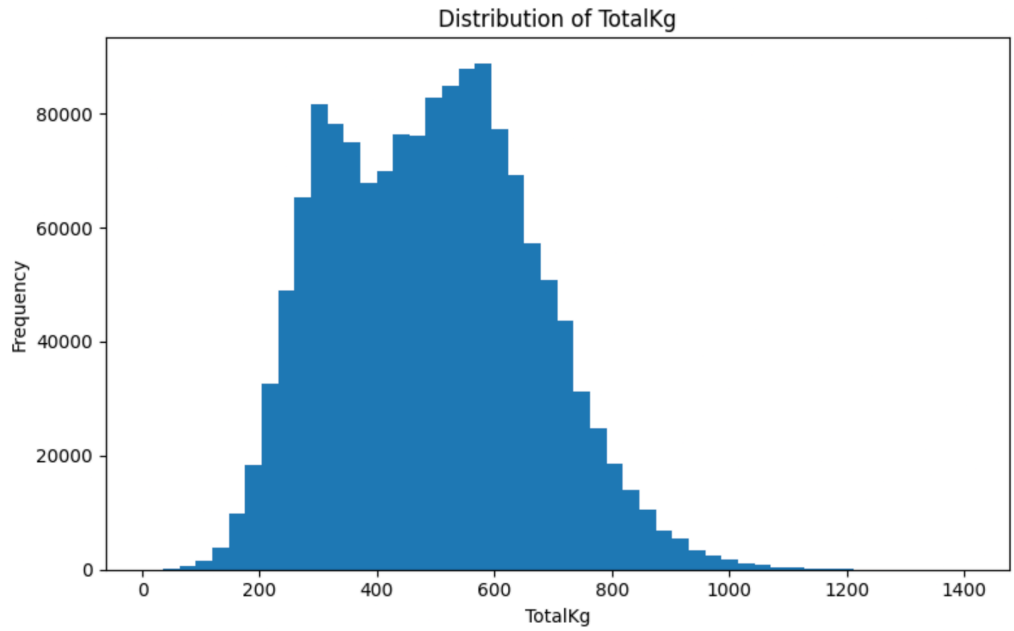

Figure 1. Distribution of TotalKg

This histogram shows the distribution of competition total weight lifted in the EDA subset. It is useful for understanding the spread of the target variable and whether the data are concentrated in certain performance ranges before regression modeling.

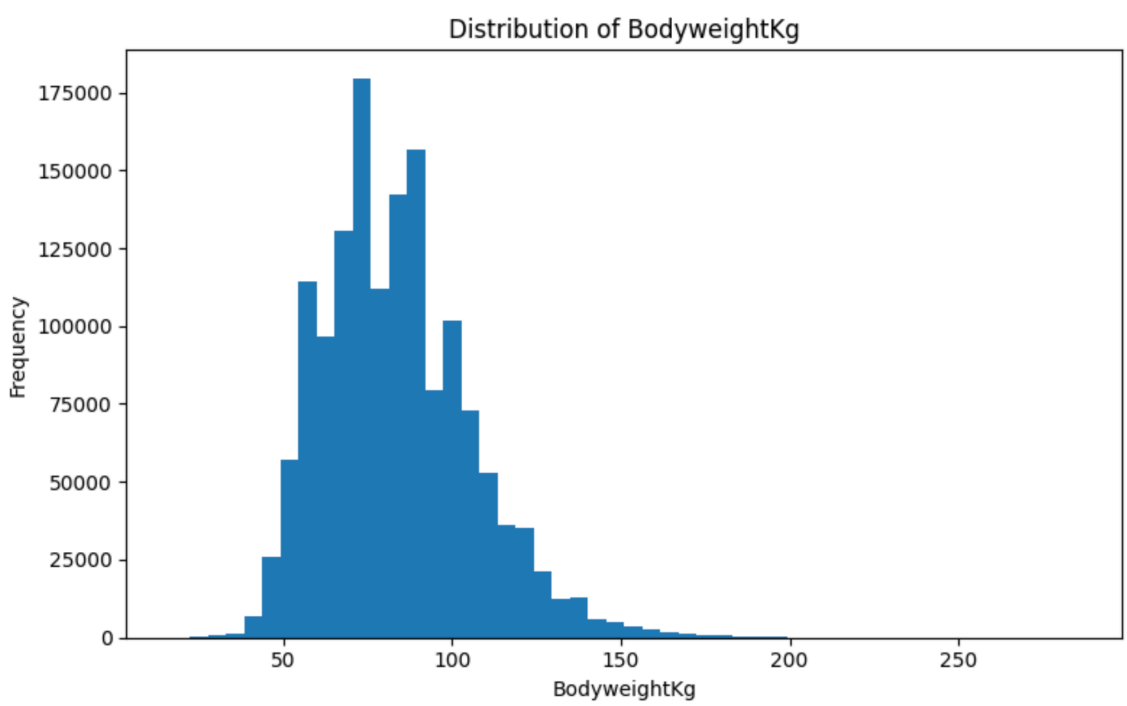

Figure 2. Distribution of BodyweightKg

This histogram shows the distribution of athlete bodyweights in the EDA subset. It helps explain the scale and variation of one of the project's most important predictors.

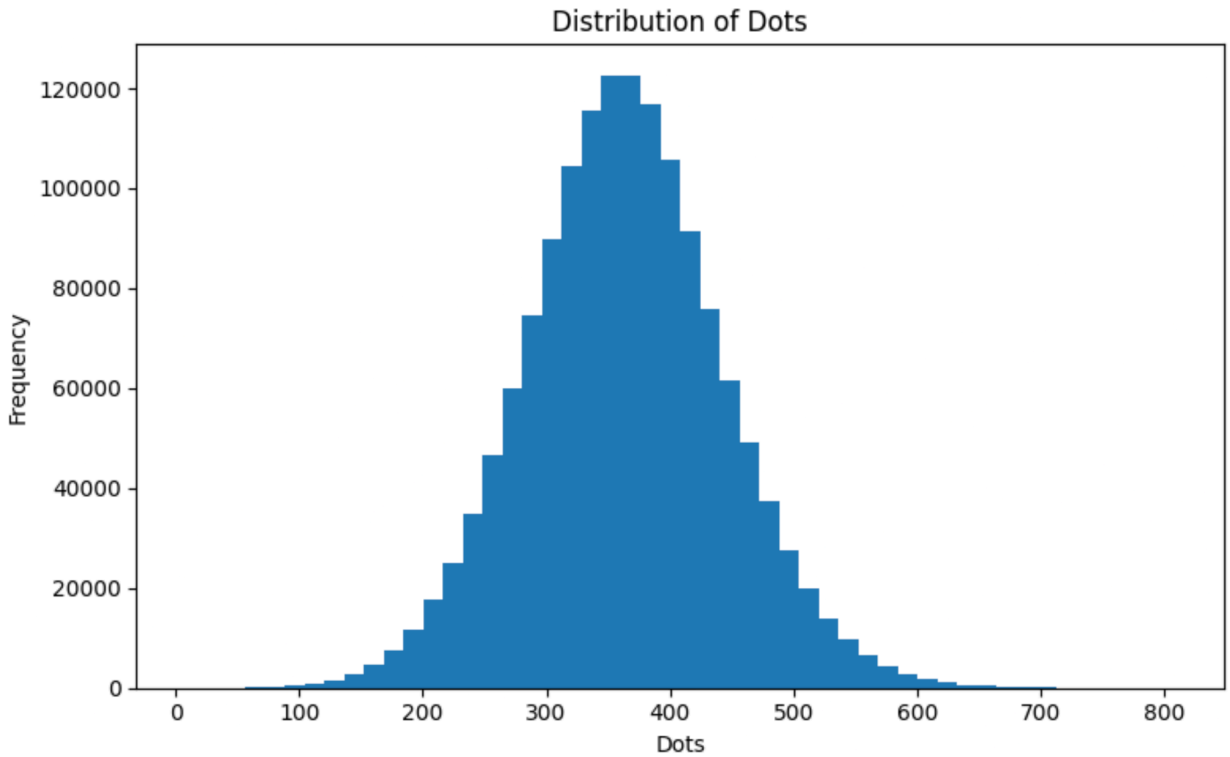

Figure 3. Distribution of DOTS
This histogram shows the distribution of DOTS scores in the EDA subset. It supports the project's secondary analysis by showing how relative-strength scores are distributed across lifters.

The descriptive statistics show that TotalKg has a mean of approximately 493.993 kg and a standard deviation of 172.113 kg, while BodyweightKg has a mean of 83.535 kg and a standard deviation of 22.012 kg. The mean age in the usable EDA subset is 28.541 years, and the mean DOTS score is 363.436. The spread of these variables confirms that the dataset contains substantial variation, which is useful for regression modeling and for comparison across athlete categories.

4.2 Category counts

As part of EDA, counts by sex and equipment were added to the analysis summary so that subgroup means would not be interpreted without context. This matters because mean performance values can be misleading if one subgroup is extremely small. Including counts strengthens the interpretation of categorical comparisons and makes the EDA more consistent with the expectation that qualitative variables should be summarized with counts and related summaries, not just with averages.

4.3 Mean total by sex

The mean TotalKg by sex shows a clear difference across sex categories: approximately 569.729 kg for M, 322.942 kg for F, and 315.467 kg for Mx. This suggests that sex category is strongly associated with absolute total lifted in the cleaned competition data. Since sex is also a standard performance characteristic in the literature, this pattern is not surprising. Huebner and Perperoglou paper found that sex-related performance differences vary across body masses and ages, and they report that "body weight at younger ages has less impact on performance compared to older ages," which supports the idea that sex and body mass both matter when analyzing strength performance (Huebner & Perperoglou, 2020).

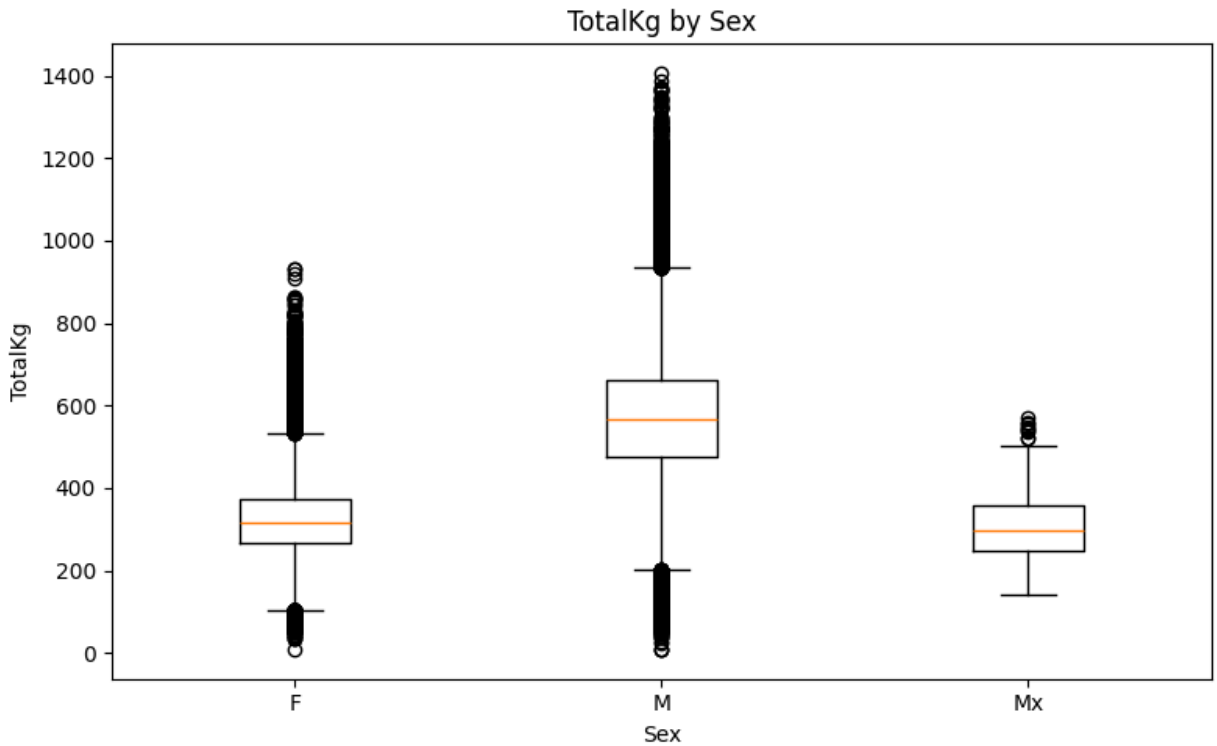

Figure 4. TotalKg by Sex
This boxplot compares the distribution of competition totals across sex categories. It helps show not only differences in average performance, but also the spread and overlap of totals across groups, supporting the decision to keep Sex in the main model

4.4 Mean total by equipment

Mean TotalKg also varies noticeably across equipment classes. In the EDA subset, Straps had the highest mean total, followed by Multi-ply, Single-ply, Wraps, Raw, and Unlimited meaning that this indicates that the equipment category is associated with differences in absolute total lifted, which supports its inclusion in the main regression model. However, these results should be interpreted carefully, especially for smaller categories, which is why the added category counts are useful. Equipment category is also meaningful in the OpenPowerlifting documentation, which distinguishes between Raw, Wraps, Single-ply, Multi-ply, Unlimited, and Straps, and explicitly notes that category labels reflect allowed equipment rather than necessarily the exact gear worn.

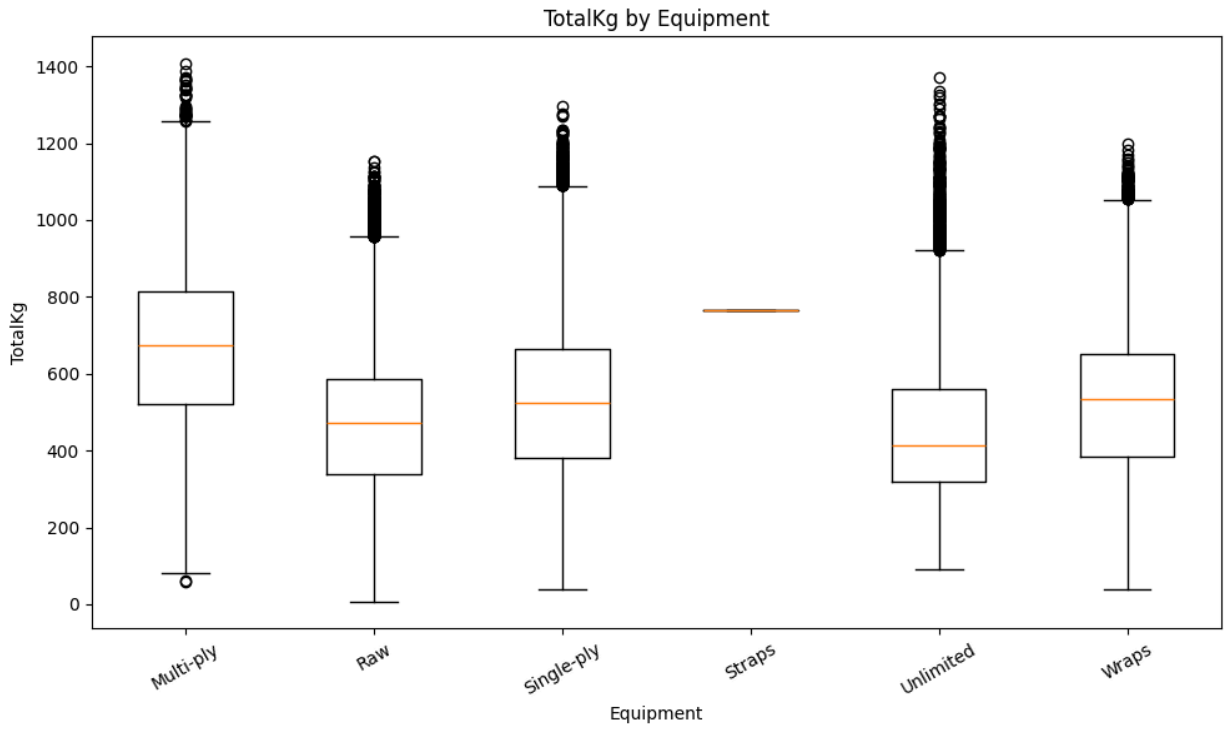

Figure 5. TotalKg by Equipment

This boxplot compares the distribution of competition totals across equipment categories. It shows that the equipment category is associated with meaningful differences in absolute lifting performance, supporting its inclusion as a predictor.

4.5 Mean DOTS by sex and equipment

DOTS was retained for secondary analysis rather than main prediction, and its grouped means provide a useful contrast with absolute strength. The mean DOTS score by sex is approximately 375.448 for M, 336.339 for F, and 218.023 for Mx. By equipment, mean DOTS is highest for Straps and Multi-ply, followed by Single-ply, Unlimited, Wraps, and Raw. These summaries show that the same factors associated with high totals are not necessarily identical in magnitude when performance is viewed through a relative-strength metric. This supports the decision to keep DOTS in the project for interpretation while avoiding it as the main target. OpenPowerlifting describes DOTS as a score that is "very similar to" Wilks but uses updated coefficients and was built using drug-tested raw data.

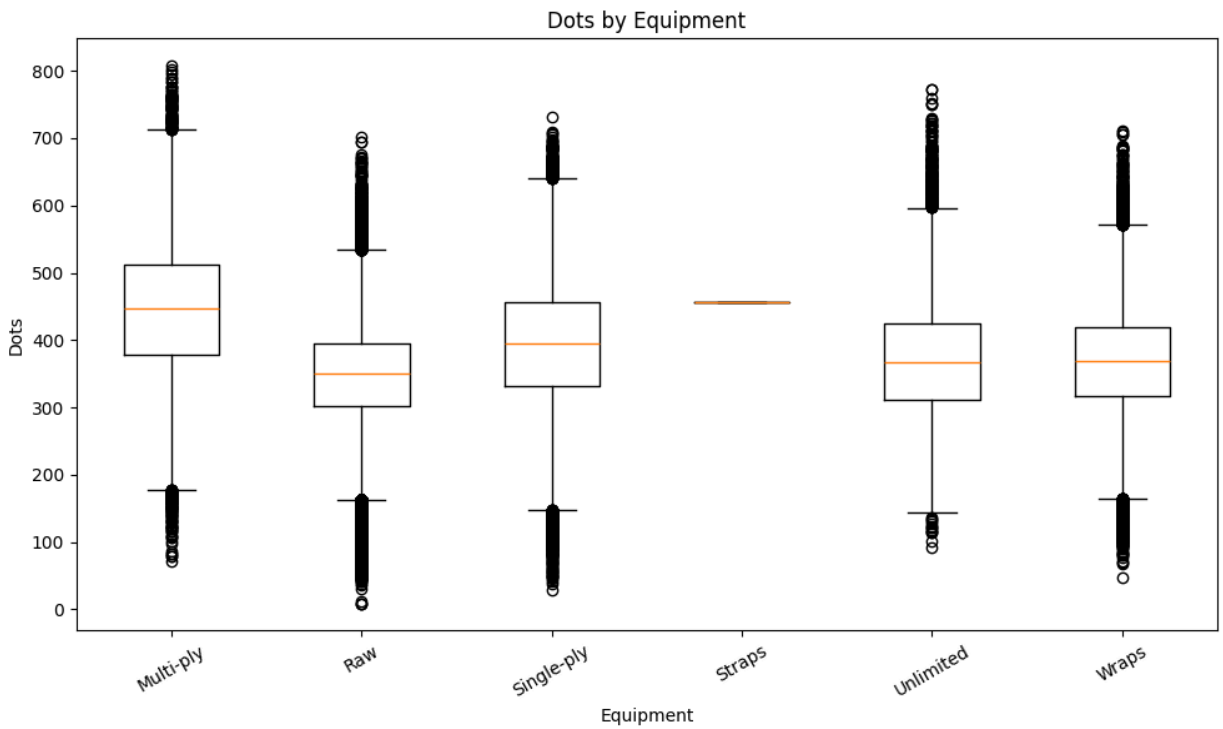

Figure 6. DOTS by Equipment

This boxplot compares the distribution of DOTS scores across equipment categories. It is important because it supports the project's secondary analysis of relative strength and shows that patterns in DOTS do not perfectly match patterns in absolute total.

4.6 Correlation matrix (non-visual insight)

The correlation matrix provides a key non-visual insight required by the rubric. In our EDA subset, TotalKg is moderately to strongly correlated with BodyweightKg (0.657) and DOTS (0.781), while its correlation with Age is much weaker (0.041). BodyweightKg and DOTS are only modestly correlated (0.198), and Age has almost no linear relationship with DOTS (-0.019). This suggests that bodyweight is an important predictor of absolute total, while age may add only a small linear contribution once other factors are taken into account. These patterns helped justify the final variable set and the use of a simple interpretable regression model.

Correlation matrix for key numerical variables				
Variable	TotalKG	BodyweightKG	Age	DOTS
TotalKG	1.000	0.657	0.041	0.781
BodyWeightKG	0.657	1.000	0.157	0.198
Age	0.041	0.157	1.000	-0.019
DOTS	0.781	0.198	-0.019	1.000

Table 4. Correlation matrix for key numerical variables
This table shows the pairwise linear relationships among the main numerical variables used in EDA. It serves as the required non-visual insight and helps justify the choice of predictors used in the final model.

4.7 Visualizations

The project includes several visualizations produced from the cleaned dataset, including histograms of numerical variables, boxplots by sex and equipment, and scatterplots for key relationships. These figures help illustrate distributions, central tendency, subgroup differences, and relationships among variables. This approach strongly emphasizes readable and well-labeled plots for both categorical and numerical data, as well as the use of visualizations to support later modeling decisions. In the final submission, each figure should be numbered and briefly interpreted in the report.

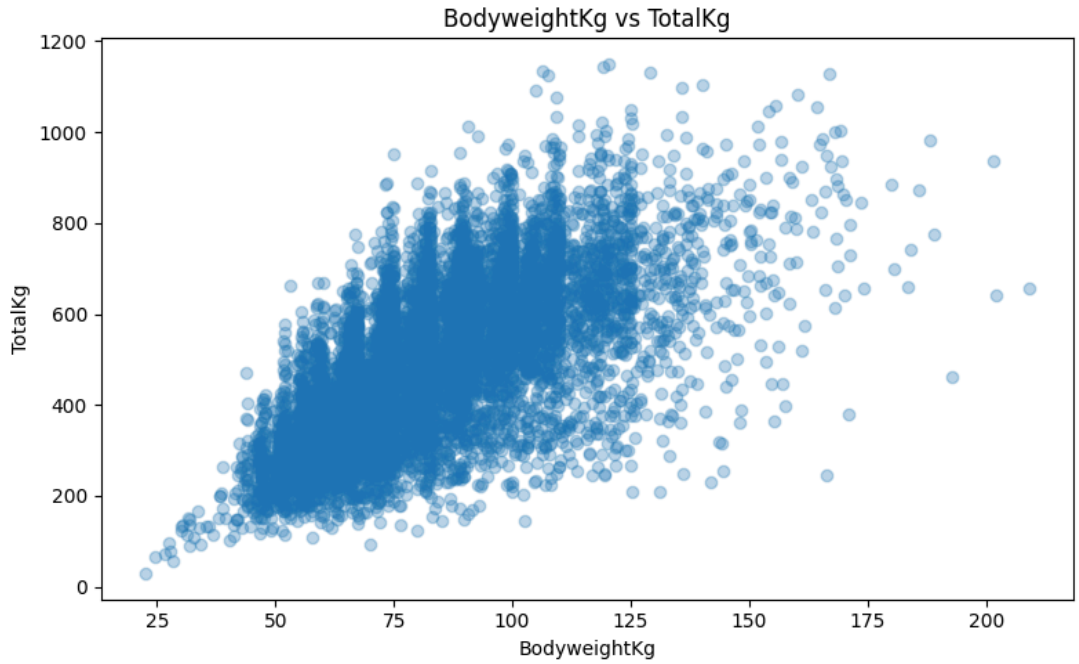

Figure 7. BodyweightKg vs TotalKg This scatterplot shows the relationship between bodyweight and competition total in a sample of the EDA dataset. It is one of the most important relationship plots in the report because it visually supports the positive association between bodyweight and absolute strength.

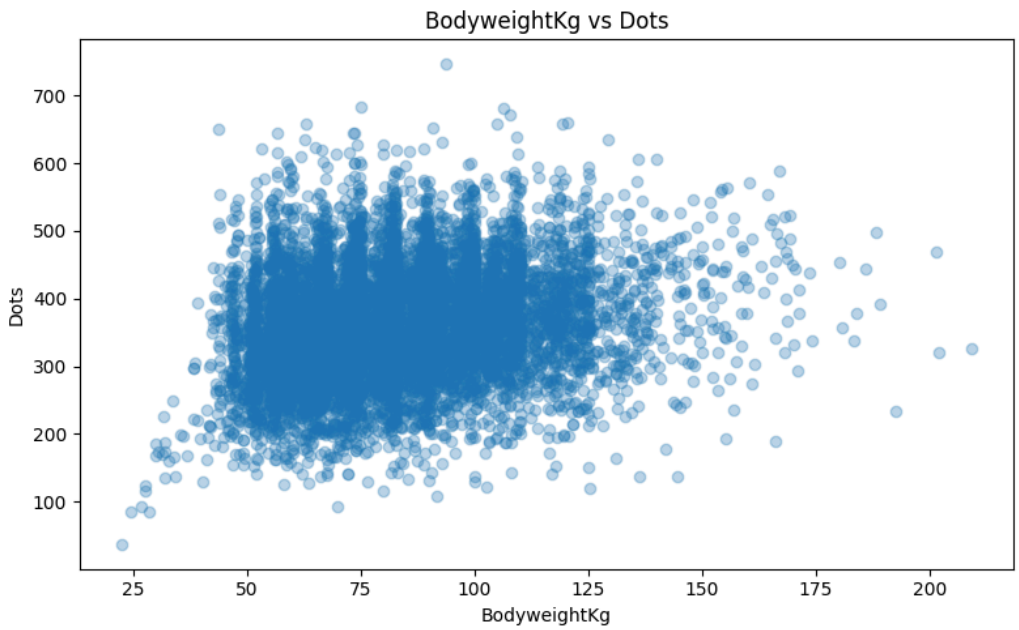

Figure 8. BodyweightKg vs DOTS This scatterplot shows the relationship between bodyweight and DOTS in a sample of the EDA dataset. It is useful because it contrasts with the previous plot and helps show that bodyweight relates much more strongly to absolute total than to relative-strength scoring

Overall, the EDA suggests that BodyweightKg, Sex, and Equipment are meaningfully associated with TotalKg, while Age is weaker but still worth considering because of its conceptual relevance and because the dataset remains very large after missing-age rows are removed. The secondary analysis using DOTS also confirms that absolute and relative performance are related but not identical, which enriches the interpretation of the final model.

5. Modeling

The main modeling task in this project is a multiple linear regression problem. This is an appropriate model choice because the dependent variable, TotalKg, is numerical and continuous, and the project aims to estimate how it changes with a small set of interpretable predictors.

final modeling specification and rationale		
Model component	Final choice	Reason
Modeling approach	Multiple linear regression	The target variable, TotalKg, is numerical and continuous, so regression is an appropriate modeling choice.
Dependent variable	TotalKg	This is the main direct measure of competition output and matches the revised research question
Predictor 1	BodyweightKg	Bodyweight is directly related to lifting performance and showed a meaningful relationship with total during eda
Predictor 2	Age	Age is conceptually relevant to performance and was kept despite missingness because the remaining usable sample was still very large
Predictor 3	Sex	Sex showed clear differences in grouped eda summaries and is a meaningful athlete characteristic
Predictor 4	Equipment	Equipment category captures meaningful competition differences and showed visible differences in total and DOTS during EDA

Variable kept for secondary analysis only	DOTS	DOTS was used to compare relative strength with absolute strength, but it was not used as the main regression target
Variables excluded from main model	Event, Tested, Division, WeightClassKG	these variables were either used mainly for filtering, had missingness, overlapped with simpler predictors or added unnecessary complexity relative to the core research question
Training method	80/20 train-test split	This creates a sizable unseen test set for evaluating generalization, which matches the project rubric
Encoding	One-hot encoding for 'Sex' and 'Equipment'	Linear regression requires numerical inputs so categorical predictors had to be encoded
Numerical transformation	z-score standardization for BodyweightKg and Age	Standardization placed numerical predictors on comparable scales and supported more stable interpretation of coefficients

Table 5. Final modeling specification and rationale This table summarizes the final regression model, including the dependent variable, predictors, excluded variables, train-test strategy, and preprocessing choices that were necessary for fitting the model.

The dependent variable for the main model is $Y = TotalKG$
The independent variables are:

- standardized BodyweightKg
- standardized Age
- encoded Sex
- encoded Equipment

These variables were selected because they align with the research question and were supported by EDA findings. BodyweightKg was kept because it is directly relevant to lifting total and had no missing values after cleaning. Sex was kept because grouped means and prior literature suggest clear performance differences across sex categories. Equipment was kept because different competition equipment categories are associated with different absolute totals and form a natural competition characteristic. Age was kept despite large missingness because it was part of the intended research question, is conceptually relevant to performance, and still leaves a very large usable sample after row removal. DOTS was excluded from the main model because it is a secondary interpretation variable rather than the main target. Event was used mainly for filtering and comparability, while Tested, Division, and WeightClassKg were left out to preserve interpretability and avoid unnecessary complexity.

This choice is also defensible in light of the literature. Ferrari et al. recently developed a machine-learning-based linear regression model for powerlifting prediction using variables such as "sex, age,

weight," which is very close to the simplified predictor family used in this project (Silverthorne et al., 2025). Huebner and Perperoglou also showed that body mass and sex differences matter in strength performance, while competition-oriented research has shown that competition factors can affect success in powerlifting (Huebner & Perperoglou, 2020). Together, this literature supports the inclusion of bodyweight, age, sex, and competition characteristics in a performance model.

The model was fit using an 80/20 train-test split with random_state =42. The model-ready dataset contained 1,470,898 rows, of which 1,176,718 were used for training and 294,180 were used for testing. This is a sizable and unseen test set, satisfying the rubric's requirement for generalization testing.

6. Model SUmmary

The fitted multiple linear regression model can be written as:

$$\widehat{TotalKG} = 490.704 + 79.486(BodyweightKg_z) - 3.609(Age_z) + 165.698(Sex_M) - 48.659(Sex_Mx) - 130.820(Equipment_Raw) - 74.888(Equipment_Single - ply) + 21.352(Equipment_Straps) - 101.759(Equipment_Unlimited) - 107.847(Equipment_Wraps)$$

where BodyweightKg_z and Age_z are z-scored versions of bodyweight and age, and the omitted categories act as the reference categories for the encoded dummy variables. The intercept is 490.704, and the coefficients above are rounded to three decimal places as required by the rubric.

Estimated coefficients for the final multiple linear regression model		
Predictor	Coefficient	Interpretation in the model
Intercept	490.704	The predicted total for the reference categories when standardized numerical predictors are equal to 0
BodyweightKg_z	79.486	A one standard deviation increase in bodyweight is associated with an increase of about 79.486 kg in predicted total, holding the other predictors constant
Age_z	-3.609	A one standard deviation increase in age is associated with a decrease of about 3.609 kg in predicted total, holding the other predictors constant

Sex_M	165.698	Belonging to category M is associated with a 165.698 kg higher predicted total than the omitted sex reference category, holding the other predictors constant
Sex_Mx	-48.659	Belonging to category Mx is associated with a 48.659 kg lower predicted total than the omitted sex reference category, holding the other predictors constant
Equipment_Raw	-130.820	Being in the Raw category is associated with a 130.820 kg lower predicted total than the omitted equipment reference category, holding the other predictors constant
Equipment_Single-ply	-74.888	Being in the Single-ply category is associated with a 74.888 kg lower predicted total than the omitted equipment reference category, holding the other predictors constant
Equipment_Straps	21.352	Being in the Straps category is associated with a 21.352 kg higher predicted total than the omitted equipment reference category, holding the other predictors constant
Equipment_Unlimited	-101.759	Being in the Unlimited category is associated with a 101.759 kg lower predicted total than the omitted equipment reference category, holding the other predictors constant
Equipment_Wraps	-107.847	Being in the Wraps category is associated with a 107.847 kg lower predicted total than the omitted equipment reference category, holding the other predictors constant

Table 6. Estimated coefficients for the final multiple linear regression model this table presents the fitted coefficients of the main regression model. It allows the model to be interpreted directly and supports discussion of which predictors are most strongly associated with competition total. Note that BodyweightKg and Age were standardized using z-scores before fitting the model and that the sex and equipment coefficients should therefore be interpreted as being relative to the omitted reference categories created by one-hot encoding

The most immediately interpretable coefficient is for standardized BodyweightKg, which is positive (79.486), indicating that heavier bodyweight is associated with a higher predicted total after controlling for the other variables. The coefficient for standardized Age is negative but much smaller in magnitude (-3.609), suggesting a relatively weak linear contribution compared with the other variables. The encoded sex and equipment coefficients reflect category-specific differences relative to the omitted baseline categories. These coefficients should be interpreted as associations in an observational sports dataset, not as direct causal effects. There is an important and explicit distinction between correlation and causation, so this caution is important.

7. Model Evaluation

The final model was evaluated on both the training and testing subsets using two regression metrics: RMSE and R^2. This satisfies the rubric requirement that at least two distinct and appropriate metrics be used. For the training set, the model achieved an RMSE of 104.186 and an R^2 of 0.6333. For the test set, the model achieved an RMSE of 104.421 and an R^2 of 0.6328.

Training and test performance of the final regression model			
Metric	Training set	Test Set	Interpretation
RMSE	104.186	104.421	The models root mean square prediction error (RMSE) is a little over 100 kg on both the training and test sets, indicating moderate prediction error relative to the scale of TotalKg
R^2	0.633	0.633	The model explains about 63.3% of the variation in competition total on both the training and test sets
Numbers of Rows	1,176,718	294,180	The model was evaluated on a very large training set and a sizable unseen test set
Generalization comparison	Very similar	Very similar	The close agreement between training and test performance suggests that the model generalizes consistently and is not obviously overfitting

Table 7. Training and test performance of the final regression model This table summarizes the model's performance on both the training and unseen test sets. It shows that the model explains a substantial portion of the variation in TotalKg and that its performance is very similar across the two subsets, which supports its generalizability

The test RMSE of about 104.4 kg means that, on average in root-mean-square terms, the model's predictions differ from actual totals by a little over 100 kilograms. Since TotalKg values span a wide range in the dataset, this level of error is not trivial, but it is also not surprising for a deliberately simple model built from only four conceptual predictors. The R^2 value of about 0.633 means that the model explains roughly 63.3% of the variation in competition total within the test set. This indicates moderate predictive strength: the model captures a substantial amount of the performance variation, but there remains a large amount of unexplained variation likely due to factors not included in the dataset, such as training history, lift-specific strengths, federation characteristics, and competition context.

A particularly important result is that the training and test metrics are extremely close. The training RMSE and test RMSE differ by less than 0.3 kg, and the training and test R^2 values differ by less than 0.001. This suggests that the model generalizes consistently and is not obviously overfitting the training data. Given the very large sample size and the simplicity of the model, this stability is a clear strength. By comparing training and test performance to assess likely real-world behavior, the current model performs well on that end.

8. Interpretation of Results

Overall, the project was reasonably successful in modeling the target variable. Using only BodyweightKg, Age, Sex, and Equipment, the model explained about 63% of the variance in competition total and generalized well from training to test data. This means that the project's small, interpretable predictor set captures an important part of competitive powerlifting performance, even though it does not account for everything that influences total lifted.

The strongest practical insight from the model and the EDA is that bodyweight is strongly associated with absolute total lifted. This is consistent with the EDA correlation of 0.657 between BodyweightKg and TotalKg, as well as the large positive standardized coefficient in the regression model. The project's secondary use of DOTS is helpful here: while heavier bodyweight tends to be associated with higher totals, bodyweight is much less strongly related to DOTS than to total, which suggests that relative-strength interpretation can differ substantially from absolute-strength interpretation. This is one of the main reasons DOTS was retained in the project. Ferland et al. studied the efficiency of scoring formulas for "comparing powerlifting performances," which reinforces the usefulness of including a relative-strength metric in discussion even when the main model predicts total lifted (Ferland et al., 2020).

The model also suggests that sex category and equipment category are important predictors of total. This is consistent with the grouped means found in EDA and with sport-science literature that identifies meaningful sex- and body-mass-related differences in performance. Huebner and Perperoglou found that the sex gap in performance changes with age and body mass, while Ferrari et al. used sex, age, and weight in a modern predictive framework for powerlifting performance. The current project's results are therefore plausible in the context of recent literature (Huebner & Perperoglou, 2020) (Ferrari et al., 2025).

One strength of the model is that it is simple and interpretable. It uses a small number of meaningful predictors, can be written explicitly as an equation, and yields consistent performance on unseen data. This makes it well suited, emphasizing understandable data science workflows and the interpretation of results, not merely black-box prediction. Another strength is the very large usable sample size even after age-based filtering, which helps make the train/test comparison stable.

One shortcoming of the model is that it does not include many performance-relevant variables that could plausibly improve prediction. It does not account for training age, coaching quality, injury status, federation effects, meet-level conditions, squat-bench-deadlift composition, or prior competition history. It also loses a large number of rows due to missing age, and that missingness may not be random. In addition, the project is based on observational competition data, which means coefficients should be interpreted as associations rather than direct causal effects.

The secondary DOTS analysis adds value because it prevents the report from treating absolute total as the only meaningful perspective on strength. OpenPowerlifting explains that DOTS was designed as an updated relative-strength scoring system, and the literature on formula efficiency shows that different scoring systems are relevant when comparing lifters across categories (Ferland et al., 2020). Including DOTS in EDA and interpretation therefore strengthens the project conceptually even though it is not the main prediction target.

9. Conclusion and Recommendations

This project examined whether an athlete's competition total could be predicted using a small set of athlete and competition characteristics from the OpenPowerlifting database. After data cleaning, variable selection, EDA, preprocessing, and model fitting, the final multiple linear regression model achieved a test R^2 of approximately 0.633 and a test RMSE of approximately 104.4 kg. These results show that BodyweightKg, Age, Sex, and Equipment provide a useful but incomplete picture of absolute competition performance. The project also showed that DOTS can be used meaningfully in secondary analysis to contrast relative and absolute strength.

The project therefore met its main goal as it built a working, interpretable predictive model from real-world public data and used that model to answer a specific performance question.

10. Advice for future researchers (Bonus Part)

Future researchers working on a similar project should consider expanding the predictor set beyond the small core used here. Variables such as prior meet history, lift-specific best attempts, federation, meet location, drug-tested status, competition experience, and division-specific context could all improve predictive performance if handled carefully. Ferrari et al. found value in including variables such as sex, age, weight, initial strength, and competition history in prediction models, suggesting that a richer but still interpretable feature set could improve this project in future work (Silverthorne et al., 2025) (Ferrari et al., 2025).

A second recommendation is to examine missingness more directly, especially for age. In this project, age was kept because it was conceptually important and still left a very large usable sample. However, future work could compare models with and without age, evaluate imputation strategies, or test whether dropping missing-age rows introduces bias. That would make the modeling decisions more rigorous and could improve generalizability.

A third recommendation is to explore more targeted modeling strategies. Separate models by sex category or equipment category may fit better than one pooled model, and interaction terms may capture differences that a simple additive model misses. Researchers could also compare linear regression with other interpretable approaches and evaluate whether the added complexity materially improves out-of-sample performance. Since recent literature notes an "exponential increase" in powerlifting research and continuing interest in performance prediction, there is clear room for more detailed and sport-specific modeling work (Silverthorne et al., 2025) (Ferrari et al., 2025).

If this project were repeated, the main change I would make would be to build a comparison between the current simple model and at least one richer but still explainable alternative, while keeping the report focused on interpretability. I would also formalize subgroup analysis more carefully, especially for equipment classes and sex categories, and include a more explicit treatment of the relationship between total-based and relative-strength-based measures.

References & Works Cited

- Silverthorne, K., Morrison, M., Cowley, N., Munteanu, G., Creaby, M. W., Timmins, R. G., Chiang, C.-Y., & Weakley, J. (2025, November). *The Applied Sport Science and Medicine of Powerlifting and Para Powerlifting: A systematic scoping review with recommendations for Future Research*. Sports medicine (Auckland, N.Z.). https://pmc.ncbi.nlm.nih.gov/articles/PMC12559058/
- Open Powerlifting. 2026. Open Powerlifting Dataset. Bulk CSV download. Revision bc2ad218. Updated April 3, 2026. https://openpowerlifting.gitlab.io/opl-csv/bulk-csv.html
- Ahmad, A., & David, E. (2026). Predicting-total-powerlifting-weights [Source code]. GitHub. https://github.com/Mehrgarithm/predicting-total-powerlifting-weights

- Ferrari, L., Bochicchio, G., Bottari, A., Lucertini, F., & Pogliaghi, S. (2025). Predicting future performance in powerlifting: A machine learning approach. *Sports Medicine - Open*, *11*(1). https://doi.org/10.1186/s40798-025-00903-z
- Huebner, M., & Perperoglou, A. (2020, September 3). *Sex differences and impact of body mass on performance from childhood to senior athletes in olympic weightlifting*. PloS one. https://pmc.ncbi.nlm.nih.gov/articles/PMC7470362/
- Ferland, P.-M., Allard, M.-O., & Comtois, A. S. (2020, September 1). *Efficiency of the Wilks and IPF formulas at comparing maximal strength regardless of bodyweight through analysis of the Open Powerlifting Database*. International journal of exercise science. https://pmc.ncbi.nlm.nih.gov/articles/PMC7523908/
-